Haptic feeback controllers
Sony has developed personalised haptic feedback in its DualSense contollers.
https://dgit.in/jan22-73

Epilepsy detection
Chennai-based startup Neurostellar develops 3D printed headsets to detect epilepsy.
https://dgit.in/jan22-74

"Entry-level" Graphics Cards

Where are they?

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Ya'll must think I'm dead-ass stupid to boo something that everyone has been eagerly waiting for so long but hear me out first. Yes, I know AMD just announced the Radeon RX 6500 XT for $199 and NVIDIA also announced the RTX 3050 for $259 at CES this year but do you really think that's what it's going to sell for? Alright, everyone knows that it's literally cheaper to get a current gen gaming console rather than building a PC and that's been the case for close to two years now. But you really need to see how brands are dealing with the new GPU launches.

Side note - I know some of you are surprised that I game and you might be wondering if I even have the reflexes to even play any of the video games coming out these days. I have two things to say to that. The first being that it's not just FPS games that come out, there are a lot many other genres that one can enjoy. Two, you bet your sweet rear-end that I'll defeat you fair and square in any of the games that I play.

Because of the whole chip shortage thingy that is still to subside, no entry-level graphics card will ever be sold at entry level prices. Industry leaders have said that the chip shortage problems will continue well into 2024 so you've got a long time to wait before things normalise. One industry might recover a little earlier than another but it's safe to say that you'll have to ride out another year of this madness.

With that said, let's look at how brands are contributing to the problem. If you were to look at an entry-level or mid-range graphics card before the chip shortage happened, what would you see? A smallish PCB with a pretty lean heatsink and probably one cooling fan. I say probably because the absolute bottom of the stack cards just had a passive heatsink and no fans. Putting even one fan would make the card more expensive to manufacture and the competition in the entry-level segment is ridiculous. Pricing your graphics card even a hundred Rupees higher than the average meant that you'd sell barely anything.

And now, look at what the recently announced entry-level cards look like. Every graphics card has at least one fan. The average number of fans from the graphics cards showcased during the RTX 3050 presentation is 2.25. More than two, you ask? Yes, because there are RTX 3050 graphics cards with THREE FANS! And those three fans sit on a massive PCB. Imagine this, the performance difference between a single-fan low-rung RTX 3050 and a fully juiced-up triple-fan RTX 3050 is going to be within the margin of error in most benchmarks. Manufacturers are no longer competing with each other. It's not just NVIDIA. AMD is no better with their RX 6500 XT. The GPU on these cards can be easily cooled with a moderately sized heatsink and one fan. The only reason manufacturers are juicing up these entry-level graphics cards with massive sized heatsinks coupled with three 90-120 mm fans is simply because THEY CAN. The manufacturers are no longer competing with each other for market share because whatever they make, will sell. This is just them testing the waters and seeing how much more money they can extract for all of these graphics cards. Consumers, out of desperation, will continue to purchase these graphics cards and manufacturers are pricing these entry-level cards way higher than what they were historically priced at. They know that these cards will sell at 2x the prices. They know that people are helpless and will buy whatever is on sale. They know that the MSRP will never be adhered to.

When these cards do come to India, they're expected to sell at about ₹20-25K but you'll not see them for less than ₹35-40K. So much for consumer friendly entry-level graphics cards. These are anything but that. I'm going to stick to my old PC. It's definitely younger than me and I'm in no hurry to upgrade. d